

Document Control

- System Name: Application

Executive Summary

- This BRD defines requirements for ingesting customer and order CSV data into Delta tables, applying data cleansing, computing a TotalAmount, joining for an SCD Type 2 ordersummary table, and aggregating daily spend for customeraggregatespend, leveraging Delta Live Tables.
- System Name: Application

Business Objectives

- Load customer and order data from CSV to Delta tables.
- Cleanse data to remove "Null"/Null literals and duplicates.
- Calculate and enrich each order with a TotalAmount field.
- Join customer and order data to create a comprehensive ordersummary table.
- Implement SCD Type 2 to preserve customer history in ordersummary.
- Aggregate daily customer spend in customeraggregatespend.
- Provide an end-to-end automated pipeline using Delta Live Tables.

In-Scope

- Ingest customer and order CSVs into respective Delta tables.
- Execute cleansing routines on customer and order data.
- Add computed TotalAmount to orders.
- Join customer and order on CustId, producing ordersummary.
- Apply SCD Type 2 change handling for customer dimension.
- Aggregate daily spend into customeraggregatespend.
- Implementation adherence to Delta Live Tables.

Out-of-Scope

- Integrations beyond customer and order CSVs.
- Security, access, SLAs, monitoring, or alerting.
- Performance tuning or scaling configuration.
- Scheduling, orchestration, or DevOps beyond using Delta Live Tables.
- Data quality metrics outside stated cleansing.
- Dashboard, reporting, or analytics visualization.

Stakeholders

- Not specified in source requirements.

Business Requirements

BR-APP-001 — Ingest Customer Data

- Description: Ingest customer CSV from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata into Delta table 'customer'.
- Business Value: Enables structured, accessible customer data management.
- Priority: High

BR-APP-002 — Ingest Order Data

- Description: Ingest order CSV from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata into Delta table 'order'.
- Business Value: Enables structured, accessible order data management.
- Priority: High

BR-APP-003 — Data Cleansing for Customer

- Description: Remove "Null"/Null literals and duplicate records from customer data.
- Business Value: Improves data quality and reliability for downstream processes.
- Priority: High

BR-APP-004 — Data Cleansing for Order

- Description: Remove "Null"/Null literals and duplicate records from order data.
- Business Value: Ensures order data consistency and integrity.
- Priority: High

BR-APP-005 — Add Computed TotalAmount

- Description: Add computed field TotalAmount = PricePerUnit × Qty to order table.
- Business Value: Enables accurate financial analytics and aggregations.
- Priority: High

BR-APP-006 — Create Ordersummary Table

- Description: Create ordersummary table by joining customer and order data.
- Business Value: Provides a 360° view of customer order activity.
- Priority: High

BR-APP-007 — Implement SCD Type 2 for Customer in Ordersummary

- Description: Manage historical changes in customer attributes with SCD Type 2 (active/inactive records, start/end dates).

- Business Value: Preserves full customer change history for audits and analytics.
- Priority: High

BR-APP-008 — Create Customer Aggregate Spend Table

- Description: Create customeraggregatespend table with daily total spend grouped by customer Name and Date.
- Business Value: Supports business decision-making via aggregate spend analytics.
- Priority: High

BR-APP-009 — End-to-End Pipeline with Delta Live Tables

- Description: Implement all ingestion, cleansing, transformation, joining, SCD, and aggregation steps using Delta Live Tables.
- Business Value: Enables reliable, automated data processing workflows.
- Priority: High

Assumptions & Constraints

- Only requirements explicitly described are assumed.
- Implementation modality is Delta Live Tables as instructed.
- SCD Type 2 field names (StartDate, EndDate, Active) are inferred but not explicitly specified.
- No environment or scheduling requirements are defined.

Success Metrics

- 100% of customer/order ingestions without error.
- All records cleansed of "Null"/Null literals and duplicates.
- Ordersummary correctly reflects SCD Type 2 changes in customer data.
- Aggregates match the sum of TotalAmount per customer per date.
- Pipeline execution is fully automated in Delta Live Tables.

Risks & Mitigations

- Risk: Ambiguity in SCD Type 2 field implementation.
- Mitigation: Adhere to standard SCD Type 2 best practices.
- Risk: Data quality issues beyond specified cleansing steps.
- Mitigation: Limit focus to defined cleansing; document limitations.
- Risk: Lack of details for scheduling, error handling, or monitoring.
- Mitigation: Escalate these as follow-on requirements.

Approval

- Approval details not specified in source requirements.

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into 'customer' table
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into 'order' table

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Bronze	customer	CustId, Name, EmailId, Region	Ingested from
	Bronze	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Ingested from
	Silver	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	SCD Type 2 m
end	Gold	customeraggregatespend	Name, TotalAmount, Date	Aggregated sp

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Ingestion	customer CSV	customer Delta table	Direct load
2	Ingestion	order CSV	order Delta table	Direct load
3	Computed field	order	order	TotalAmount = PricePerUnit × Qty
4	Cleansing	customer	customer	Remove "Null"/Null, deduplicate
5	Cleansing	order	order	Remove "Null"/Null, deduplicate
6	Join	customer, order	ordersummary	Join on CustId
7	SCD Type 2	customer → ordersummary	ordersummary	Maintain history, active/inactive tracking
8	Aggregation	ordersummary	customeraggregatespend	sum(TotalAmount) group by Name, Date

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-001	Computation	TotalAmount = PricePerUnit × Qty	BRD requirements
BR-002	Cleansing	Remove "Null"/Null and duplicates	BRD requirements
BR-003	Join	Join customer to order on CustId	BRD requirements
BR-004	SCD Type 2	SCD2: set previous inactive, new active, update start/end dates	BRD requirements
BR-005	Aggregation	Group by Name, Date; sum TotalAmount	BRD requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customer and order CSVs	Source files present
2	Apply cleansing to both tables	Step 1 complete
3	Add computed TotalAmount to orders	Step 2 complete
4	Join customer and order to create ordersummary	Step 3 complete
5	Apply SCD Type 2 logic on ordersummary	Step 4 complete
6	Aggregate ordersummary to customeraggregatespend	Step 5 complete

Appendix F: Entity Relationship Diagram (ERD)

